Gamal Asran

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Education

Egypt-Japan University of Science and Technology *Expected Jul 2027*
B.Sc. Computer Science *Alexandria, Egypt*

Projects

NIST AI Risk Assistant — Local RAG Application GitHub

- Independently built a retrieval-augmented generation (RAG) application for question answering across three NIST AI risk publications using Qwen3-4B through Ollama.
- Indexed 447 text chunks with MiniLM embeddings and ChromaDB, preserving document and page metadata for semantic search and source inspection.
- Integrated a FastAPI backend with a Streamlit chat interface supporting streamed responses, conversation history, and retrieved source references.

Mamba-Based Single-Object Tracking — ICMTC 2026, AIC_4 GitHub

- Adapted TrackingMamba in PyTorch by resolving legacy dependency conflicts and debugging model execution for a resource-constrained tracking competition.
- Containerized GPU inference with Docker and implemented latency benchmarking against the competition's 30 ms inference budget.
- Delivered higher Intersection over Union (IoU) than the team's alternative tracker; the selected submission helped the team place 14th on the first-phase Kaggle leaderboard.

Tweet Sentiment API — Team Project GitHub

- Built a FastAPI inference service for a team-developed LSTM sentiment classifier, returning predicted labels and confidence scores.
- Integrated text preprocessing and saved Keras model artifacts into the serving pipeline and packaged the API with Docker.

Internships & Technical Training

AI Trainee — *Information Technology Institute (ITI)* *Aug–Sep 2026*

- Completed daily practical exercises in applied Python, computer vision, NLP, and RAG during a one-month training program.
- Independently developed a local-LLM document assistant as the final project, gaining practical experience with Ollama and local inference.

Technical Intern — *Orange Egypt* *Jul–Aug 2026*

- Studied Linux fundamentals using engineer-provided manuals and discussed Kubernetes concepts with a supervising DevOps engineer.
- Shadowed network operations teams, observing ticketing, network monitoring, and log-based email automation for maintenance notifications.

Machine Learning Intern — *National Telecommunication Institute (NTI)* *Aug–Sep 2025*

- Studied the mathematical foundations of machine learning models through practical application exercises.
- Applied feature engineering and hyperparameter tuning to investigate their effects on model performance.

Data Science Intern — *Digital Egypt Pioneers Initiative (DEPI)* *Oct 2024–May 2025*

- Completed practical training in Python-based data analysis followed by machine learning.
- Developed the FastAPI serving layer and Docker packaging for the team's tweet sentiment classification final project.

Leadership & Mentoring

Machine Learning Track Lead & Mentor — *EJUST CS Club* *Mar–Aug 2026*

- Led curriculum design and taught ML fundamentals and supervised learning to first- and second-year students using slides and worked examples.

Competitive Programming Mentor — *CPC* *Nov 2024–Aug 2026*

- Supported beginners in algorithms and data structures, answering questions and explaining approaches to programming problems.

Awards

3rd Place — Smart Healthcare Hackathon *Feb 2026*
Banha University

- Built a hybrid CNN–RNN model that fused ECG features for multiclass risk classification in the team's cardiac-monitoring prototype.

Technical Skills

Languages: Python, SQL, C++
AI & Data: PyTorch, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Hugging Face, LangChain, Ollama, Sentence-Transformers, ChromaDB
Development & Deployment: FastAPI, Streamlit, Docker, Docker Compose, Git, Linux